

# 1D CNN — Architecture Variant Comparison

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## Data & configuration

|                     |                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Training path 1     | /home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic                                                                                                           |
| Batch-test root     | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic                                                                                                      |
| Total trials        | 360 (train pool 324, hold-out 36)                                                                                                                                  |
| Classes             | 6: Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap |
| Sensor channels     | 164 (sequence length 90)                                                                                                                                           |
| Resample / filter   | 90 steps · Butterworth cutoff 10.0 Hz, order 4                                                                                                                     |
| Normalisation       | per_trial_zscore                                                                                                                                                   |
| CV folds · shuffle  | 4 · True                                                                                                                                                           |
| Augmentation        | on · copies=10 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout                                                               |
| Epochs · batch size | 60 · 16                                                                                                                                                            |

### Sensor grid (✓ enabled, ✗ disabled, — not present)

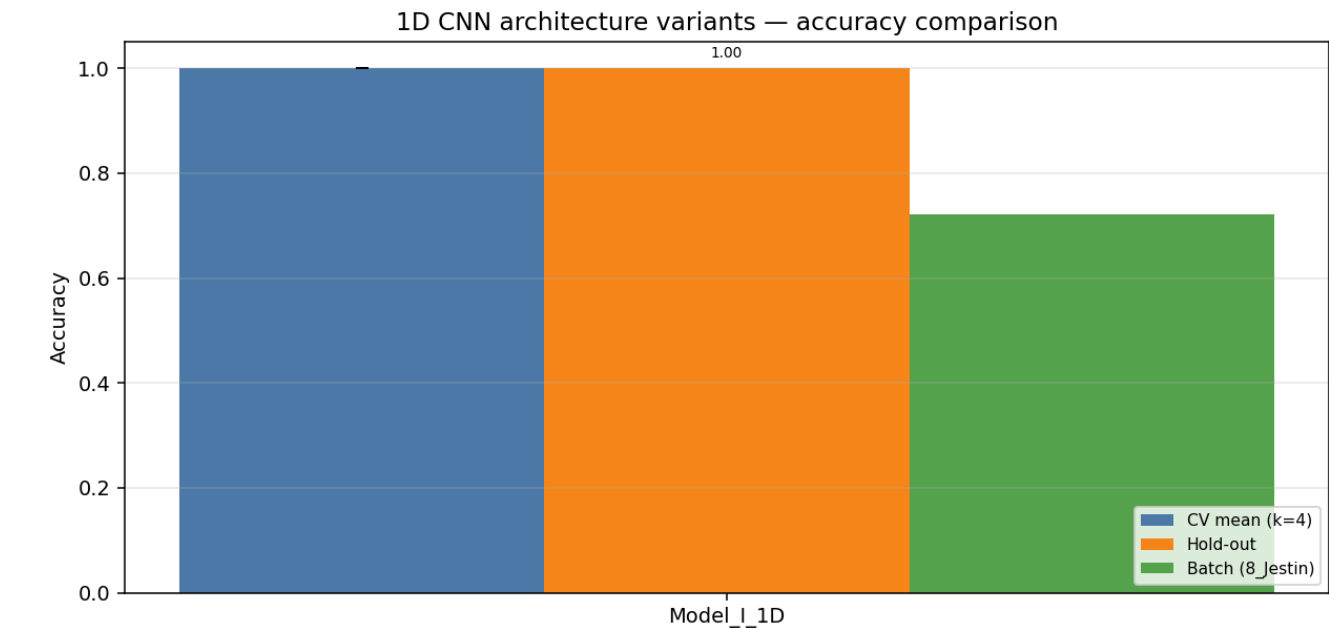
| Hand / Segment | IMU_mid | IMU_prox | Flex_MCP | Flex_PIP |
|----------------|---------|----------|----------|----------|
| left palm      | ✓       | —        | —        | —        |
| left thumb     | ✓       | ✓        | ✓        | ✓        |
| left index     | ✓       | ✓        | ✓        | ✓        |
| left middle    | ✓       | ✓        | ✓        | ✓        |
| left ring      | ✓       | ✓        | ✓        | ✓        |
| left pinky     | ✓       | ✓        | ✓        | ✓        |
| left wrist     | ✓       | —        | —        | —        |
| right palm     | ✓       | —        | —        | —        |
| right thumb    | ✓       | ✓        | ✓        | ✓        |
| right index    | ✓       | ✓        | ✓        | ✓        |
| right middle   | ✓       | ✓        | ✓        | ✓        |
| right ring     | ✓       | ✓        | ✓        | ✓        |
| right pinky    | ✓       | ✓        | ✓        | ✓        |
| right wrist    | ✓       | —        | —        | —        |

Enabled: 44 · Disabled: 0 · Modalities: YPR, Accel, Flex → 164 channels total.

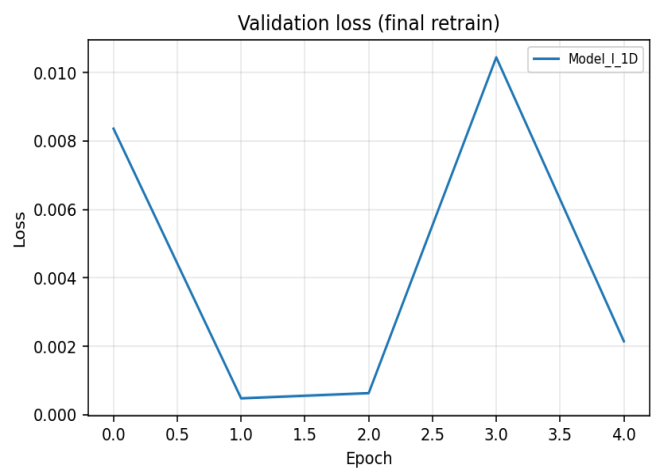
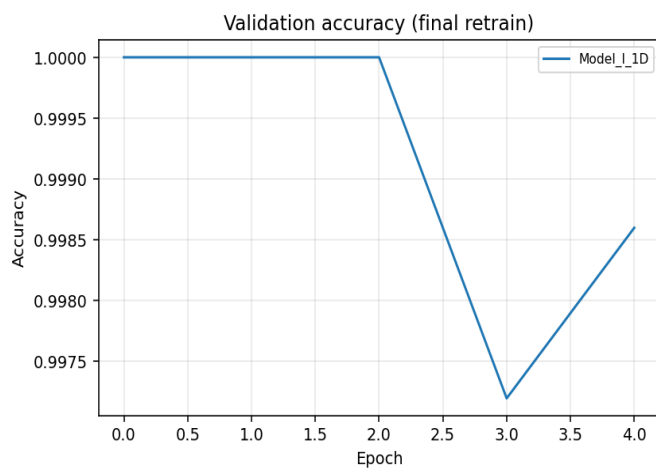
## Variant comparison (summary)

| Variant    | Params  | CV mean (k=4) | CV std | Hold-out acc | Hold-out loss | Batch acc | Batch n |
|------------|---------|---------------|--------|--------------|---------------|-----------|---------|
| Model_I_1D | 108,966 | 1.0000        | 0.0000 | 1.0000       | 0.0001        | 0.7222    | 26/36   |

Best on hold-out: **Model\_I\_1D** · Best on CV: **Model\_I\_1D**



## Training curves (final retrain on full training pool)



# Variant: Model\_I\_1D

1-D adaptation of published Model I:  
Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 108,966         |
| CV mean ± std            | 1.0000 ± 0.0000 |
| Hold-out test acc / loss | 1.0000 · 0.0001 |
| Batch-test acc           | 0.7222 (26/36)  |

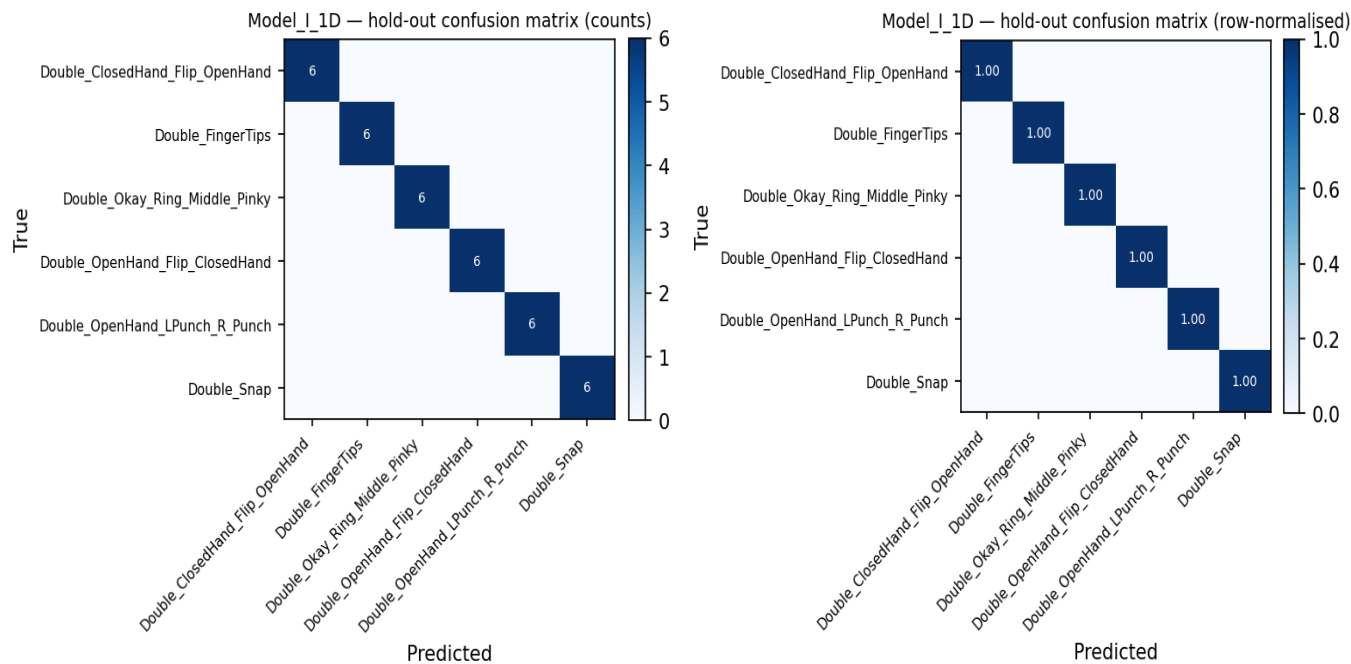
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 2673             | 81  | 1.0000  | 1.0000   | 0.9877    |
| 2    | 2673             | 81  | 1.0000  | 1.0000   | 0.9506    |
| 3    | 2673             | 81  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 2673             | 81  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

## Hold-out confusion matrix



## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 3       | 6     | 0.500    |
| Double_FingerTips               | 0       | 6     | 0.000    |
| Double_Okay_Ring_Middle_Pinky   | 6       | 6     | 1.000    |
| Double_OpenHand_Flip_ClosedHand | 6       | 6     | 1.000    |
| Double_OpenHand_LPunch_R_Punch  | 5       | 6     | 0.833    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 26      | 36    | 0.722    |

## Batch confusion matrix (8\_Jestin)

